



Autumn Examinations 2012

Exam Code(s)	1MF1, 1SD1, 2CS1
Exam(s)	Higher Diploma in Applied Science (Software Design & Development), M.Sc in Software Design & Development, BSc (Computing Studies)
Module Code(s) Module(s)	CT853, CT232 Algorithmics & Logical Methods, Methodology
Paper No. Repeat Paper	
External Examiner(s) Internal Examiner(s)	Prof. Michael O'Boyle Prof. Gerard Lyons Dr. Michael Madden Martina Fox

Instructions:

Duration	2 hours
No. of Pages	2
Discipline(s)	Information Technology
Course Co-ordinator(s)	

Requirements:

MCQ	Release to Library: Yes	☐	No	☐
Handout				
Statistical/ Log Tables				
Cambridge Tables				
Graph Paper				
Log Graph Paper				
Other Materials				

- Q1**
- Using the following methods write down(step by step) the position of each letter using the word 'KEYBOARD' when sorted alphabetically using :
 - Selection Sort
 - Merge sort'[10]
 - Which method runs fastest for an array with all keys identical, selection sort or insertion sort? [5]
 - Describe two algorithms of finding the oldest person in a room full of people. Explain which is better and why. [10]

- Q2**
- What is an algorithm and why do we study them? [5]
 - Describe the 'boundedness' property of an algorithm [5]
 - Write a algorithm that, given an array $a[]$ of N distinct integers, finds a *local maximum*: an index i such that $a[i] < a[i-1]$ and $a[i] < a[i+1]$. Your program should use $\sim \lg_2 N$ comparisons in the worst case [15]

- Q3**
- Explain the Divide & Conquer approach to algorithm design. Give an example (using pseudocode) of a Divide & Conquer algorithm to calculate X^n , where n is a positive integer [10]
 - Using Karatsuba's algorithm multiply 1234 by 5678 .Describe the algorithm explain why it is more efficient than 'standard' multiplication and give (in 'big oh' notation) its complexity [15]

- Q4** A container truck has total (weight) capacity of 7 kilo Tonnes. In a port there are five loads it can collect , each with a different weight and value:

Load	A	B	C	D	E
Weight (Kilo Tonnes)	2	5	1	1	3
Value (Millions of Euro)	11	24	7	8	20

The objective is to decide which loads to take to maximise the *Value*. Using *Dynamic Programming*, calculate which loads should be chosen. You must show your calculations in detail by writing the table of values calculated by the algorithm

[25]